# 841AR | Super Shield Nickel Conductive Coating

Broad-spectrum EMI shielding for plastic enclosures



## Description

841AR is a nickel-filled conductive acrylic coating for broad-spectrum EMI/RFI shielding of plastic electronic enclosures and assemblies.

It forms a durable conductive network that effectively attenuates both electric and magnetic field interference across a wide frequency range. The coating adheres well to most engineering plastics, including ABS, PVC, nylon, and polycarbonate. 841AR dries quickly at room temperature and can be applied by brush, spray gun, or selective coating equipment for prototype through production-scale applications.

## Key Characteristics

### Shielding Effectiveness

| | |
|---|---|
| 10 MHz to 100 MHz: | 42–55 dB |
| 100 MHz to 10 GHz: | 55–89 dB |
| 10 GHz to 18 GHz: | 68–93 dB |

### Electrical Properties

| | |
|---|---|
| Resistivity: | 4.0 × $10^{-3}$ Ω·cm |
| Surface Resistance @ 25 µm: | 1.60 Ω/sq |

### Coating Thickness & Particle Size

| | |
|---|---|
| Recommended: | 25 µm |
| Minimum achievable: | 25 µm |
| Particle Size Distribution: | D10 4 µm, D50 11.5 µm, D90 30.9 µm |




## Features & Benefits

- Durable coating with good flexibility and abrasion resistance
- Strong environmental and corrosion resistance for humid and marine environments
- Conforms easily to complex geometries and molded structures
- Safe for use on polystyrene and other delicate plastics
- Fast-drying coating suitable for brush, spray, and selective coating processes
- Ambient-temperature cure with optional heat acceleration
- UL Recognized (File # E202609)

## Applications

- Sensors and instrumentation systems
- Industrial automation and control electronics
- RF and telecommunications equipment
- Aerospace and defense electronics
- Lightweight electronic housings and subsystems
- Antenna and RF test enclosures
- Radar-absorbent surface treatment
- Automotive electronics and EV systems

## Ordering Information

| Part # | Pkg. | Net Vol. | Net Wt. |
|---|---|---|---|
| 841AR-P* | Pen | 5.0 mL | 7.57 g |
| 841AR-55ML | Bottle | 55 mL | 92.8 g |
| 841AR-340G* | Aerosol | 326 mL | 340 g |
| 841AR-900ML | Can | 850 mL | 1.43 kg |
| 841AR-3.78L | Can | 3.60 L | 6.07 kg |
| 841AR-18.9L | Pail | 18.9 L | 31.9 kg |

*See separate TDS for aerosol and pen formats.

## Physical Properties

| | |
|---|---|
| Colour | Dark Gray |
| Pencil Hardness | 3H, Hard |
| Salt Fog Resistance | Excellent |
| Service Temperature Range | -40—120 °C |

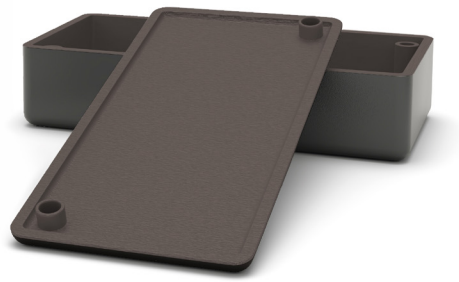
## Electrical Properties

| | |
|---|---|
| Surface Resistance @ 50 µm | 0.80 Ω/sq |
| Surface Resistance @ 25 µm | 1.60 Ω/sq |
| Resistivity | 4 x $10^{-3}$ Ω·cm |
| Magnetic Class | Ferromagnetic |

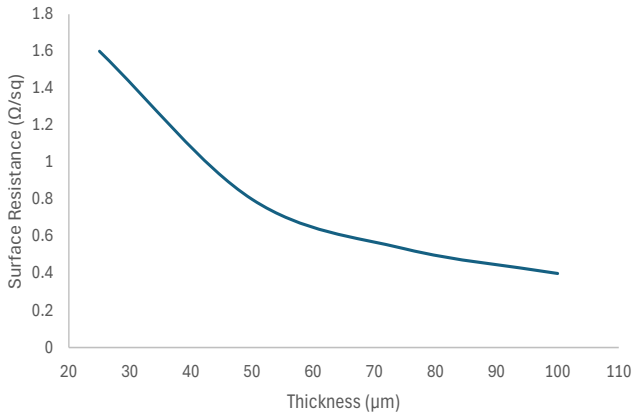
*Cured properties unless otherwise stated.*

## Cure Schedule

| | |
|---|---|
| 25°C | 24 hours |
| 65°C | 30 minutes |



## Surface Resistance by Paint Thickness




## Adhesion Performance

| | |
|---|---|
| ABS | 5B |
| Aluminum | 0B |
| Copper | 0B |
| Polycarbonate | 5B |
| Polyamide | 5B |
| Glass | 0B |
| PVC | 5B |
| FR4 | 5B |
| Stainless Steel | 2B |

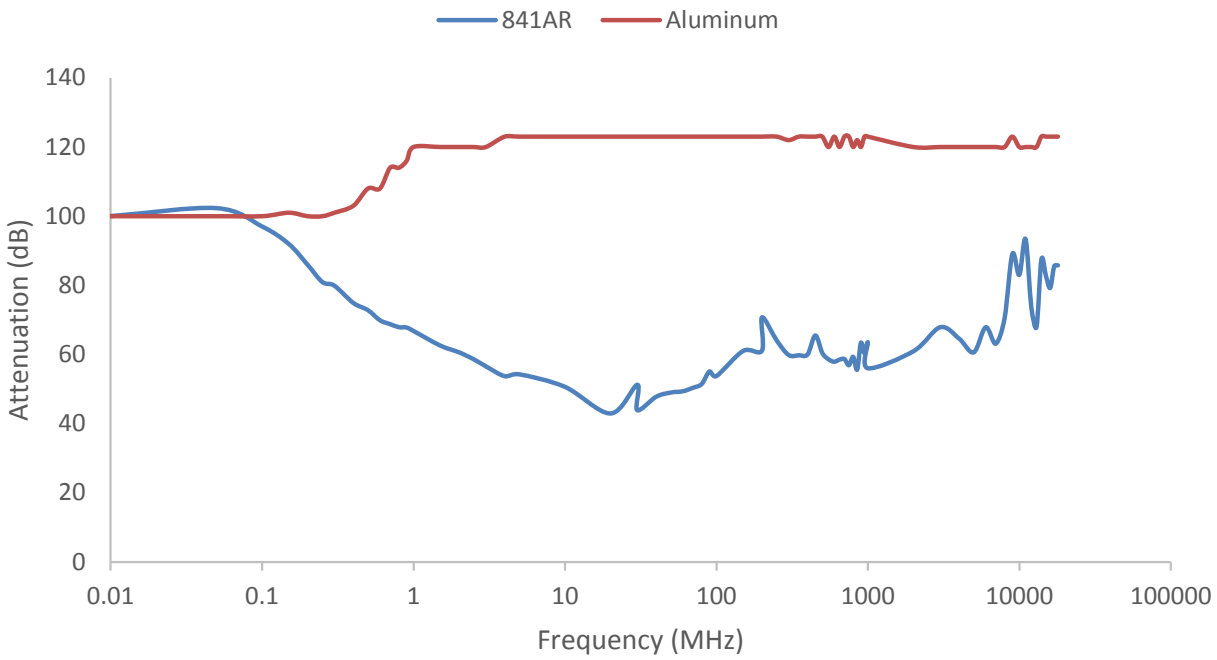
## Uncured Properties

| | |
|---|---|
| Film Thickness - Recommended | 25 µm |
| Film Thickness - Minimum | 25 µm |
| Coverage @ 25 µm | 95,570 $cm^2$/L |
| Density | 1.7 g/mL |
| Viscosity @ 23 °C | 1460 cP |
| Recoat Time | 3 min. |
| Percent Solids | 57% |
| Shelf Life | 5 y |
| VOC Content | 483 g/L |

## Shielding Effectiveness At Selected Frequencies

| Frequency | Attenuation | Note |
|---|---|---|
| 1 MHz | 67 | Lower limit of most EMC standards |
| 100 MHz | 54 | Core commercial EMC band |
| 1 GHz | 63 | Boundary between RF and Microwave |
| 3 GHz | 68 | Upper limit of most broadband shielding specs |
| 10 GHz | 83 | X-band radar, critical for aerospace and defense applications |

## Shielding Attenuation Graph




## Standards Used

| | | | |
|---|---|---|---|
| Adhesion | ASTM D3359 | Viscosity | ASTM E3116 |
| Pencil Test | ISO 15184 | Shielding Effectiveness | MIL-STD-285 |
| Salt Fog | ASTM B117 | Resistivity | MIL-STD-883J Method 5011.6 |
| Density | ASTM D1475 | | |

# Additional Resources

Scan a QR Code or click a link to access supporting documentation and product resources.

### Product Webpage

Full product information, downloads, videos and more

### SDS

Safety Data Sheet

### Application Guide for AR Conductive Coatings

Application instructions and best practices

### QC Guide for Conductive Coatings

Testing methods and quality control procedures

### Category Data Sheet for AR Conductive Coatings

Technical overview and performance summary

### EMI Shielding Coatings Catalogue

Complete solutions for EMI shielding

## Ready to Solve Your EMI Shielding Challenge?

Our application engineers are here to help you select the right materials and achieve optimal shielding performance.

- Material Selection
- EMI Troubleshooting
- Process Optimization
- Custom Solutions

### Talk to an Application Engineer

Get expert recommendations tailored to your design, materials, and requirements

**Contact Engineering**

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